Complexity Solution

Q1: O(n)

Singleloopruns n times.

Q2: O(n^2)

Twonestedloops each run n times.

Q3: O(n^2)

Iterationssum to 1+2+...+n ≈ $n^2/2$.

Q4: O(log n)

Variabledoubles every iteration.

Q5: O(log n)

Variablehalvesevery iteration.

Q6: O(n log n)

Outerloopntimes,inner loop log n times.

Q7: O(n)

ndecreases by 1 until zero.

Q8: O(log n)

Repeateddivision by 2.

Q9: O(n)

Twoseparate O(n) loops $\Rightarrow$ O(2n)=O(n).

Q10: O(n^2)

Dominatedbyquadratic loop.

Q11: O(n^3)

Threelevelsof iteration.

Q12: O(log n)

Multipliedby3each iteration.

Q13: O(n^3)

Sumofsquares grows as n^3.

Q14: $O(n)$

Recursiondepth n.

Q15: $O(\log n)$

Inputhalvedeachrecursive call.

Q16: O(2■)

$T(n)=2T(n-1)+1$.

Q17: O(2■)

NaiveFibonacci recursion.

Q18: $O(n)$

Onerecursive call per level.

Q19: $O(n)$

Linear recursion.

Q20: $O(\log n)$

Recursion on $n/2$.

Q21: $O(n)$

$T(n)=T(n-1)+1$.

Q22: $O(n^2)$

$T(n)=T(n-1)+n$.

Q23: $O(\log n)$

$T(n)=T(n/2)+1$.

Q24: O(2■)

$T(n)=2T(n-1)+1$.

Q25: $O(n)$

$T(n)=T(n/2)+n$.

Q26: $O(n \log n)$

$\Sigma \log i = O(n\log n)$.

Q27: $O(n \log n)$

$\log n \times n$.

Q28: O(n²)

$n+(n-1)+\ldots+1 = n(n+1)/2$.

Q29: O(n log n)

$n \times \log n$.

Q30: O(n)

Master theorem: $T(n)=2T(n/2)+1$.